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A Large-Scale Social Media Corpus for the Detection of Youth Depression (Project Note)

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Abstract

Social media is frequently used by youth to share their health and mental issues. Therefore, social media has become a major online resource to study the language used to express issues such as depression and self-harm which can help to identify individuals at risk of harm. Furthermore, depression and suicide are generally closely related especially that depression is the most common symptom associated with self-harm acts such as suicide. In this project, we propose to build a linguistically annotated corpus with the sentiment analysis in order to study the youth behavior through their social media discourse across the MENA region. We plan to create a large-scale dataset of users with self-reported depression messages. Several correlational analyses will be performed to understand the psycho-social-behaviors. We plan to annotate the collected corpus using a team of dedicated annotators from various Arabic countries. Moreover, we will use various natural language processing (NLP) tools and techniques to reveal the linguistic patterns and the sentiments expressed by these tweets. Finally, we will apply machine learning (ML) methods to build behavior prediction tools using the annotated corpus. We believe that the annotated corpus to will be a valuable resource to be used by linguists, sociologists, computer scientists, psychologists, policy makers, etc.

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Keywords: Social Media Analysis, Corpus Annotation, Arabic Dialects, Depression

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1. Background

Social media is frequently used by youth to share their health and mental issues. Therefore, social media has become a major online resource to study the language used to express issues such as depression and self-harm which can help to identify individuals at risk of harm. Furthermore, depression and suicide are generally closely related especially that depression is the most common symptom associated with self-harm acts such as suicide. Informed societies are new beneficiaries of big data. What starts as social online discourse could hold prime value for identifying pertinent risks and opportunities. Researchers are using information garnered from social media to provide insights to explain human behavior.

Within such a context, we plan to address the issue of depressive behavior and self-harm in the Middle East and North Africa (MENA) region through the creation and the annotation of a large-scale Arabic corpus of Tweets consisting of users discussing their depression symptoms. In recent years, depression in youth has been an issue of concern in many countries such as the U.S.A., where researchers from the Centers for Disease Control and Prevention investigated predictors of risky behaviors to find a possible treatment. They also created the Youth Risk Behavior Survey (YRBS) of adolescent health risks and health protective behaviors. This biennial survey is one of the major sources of public information related to these behaviors, and it is frequently used by law enforcement agencies to track behaviors such as sexual misbehavior and drug abuse.

In the MENA region, however, less is known about this, and there is a more unique cultural and regional context. As yet, depression risk indicators are not comprehensively analyzed or defined. Risky behaviors have the potential to become more widespread and problematic. In addition, unlike in the U.S.A, there is currently less support in the MENA region for mental health interventions, specifically for the young male population whose members might show depressive behaviors, but be less likely to seek help, in contrast to young female demographics.

In this project, we propose to build a linguistically annotated corpus with the sentiment analysis in order to study the youth behavior through their social media discourse across the MENA region. We plan to create a large-scale dataset of users with self-reported depression messages. Several correlational analyses will be performed to understand the psycho-social-behaviors. We plan to annotate the collected corpus using a team of dedicated annotators from various Arabic countries. Moreover, we will use various natural language processing (NLP) tools and techniques to reveal the linguistic patterns and the sentiments expressed by these tweets. Finally, we will apply machine learning (ML) methods to build behavior prediction tools using the annotated corpus. We believe that the annotated corpus to will be a valuable resource to be used by linguists, sociologists, computer scientists, psychologists, policy makers, etc.

2. Related Works

Several studies have investigated risky youth behavior in Western communities [1]. However, such work has usually focused on alcohol usage and sexual behavior as the main aspects of risky youth behavior [1]. Zafarani and Liu [2] and Zafarani et al. [3] explored a behavioral-modeling approach to explore connected users in social media sites according to their unique behavioral patterns. Pennebaker et al. [4], [5] developed the widely used Linguistic Inquiry and Word Count (LIWC), a text analysis tool that computes the percentage of words in a corpus according to over 80 linguistic features such as the first-person usage or the singular pronouns usage and also psychological state such as the anger or the achievement. His work is inspired on previous research confirming strong links between linguistic patterns and psychological state as found in Pennebaker et al. [4]. Risky youth behavior was also studied and measured based on online behavior. For example, Ybarra and Mitchell, [6] studied how social networking is risky for youth because of the potential for sexual solicitation and harassment. Walok et al., [7] also studied how communicating with strangers in online social networks can lead to risk in some cases. These studies again focused on the risk to youth from specific angles, including sexual harassment.

Other studies have used online social behavior of people to monitor their wellness. Specifically, De Choudhury [8], [9] used behavior of youth and Twitter users in general to detect any sign of depression. Their aim was to build a

model that detects and relies on several signs from social media behavior to predict the potential depression of some users at early stages. This work is highly relevant to the methodology we would like to use in our work here. However, our focus will not be mainly on depression prediction, but on prediction of potential behavior that can lead to risky actions by youth, specifically in the Arab world. Although no work has been published on using online social media to predict risky youth behavior in the MENA region, there have been some recent studies about using social media to study specific topics in Qatar for example. Weber [10] studied unemployment in Qatar and the challenges facing Qatari youth. Also, Abokhodair et al., [11] studied how Qatari youth use social media in general and Twitter in particular. They showed that many of the Qatari traditions are highly preserved with online social media platforms. This is actually important information that would guide us in our work in this project because it is in an arena in which no large signals of unethical or risky behavior have been detected before. This makes the mining for such signals on social media more challenging, yet more meaningful when detected.

From a behavioral science perspective, several researchers studied disorders such as the body dysmorphic disorder (BDD) which is a psychological disorder in which the person afflicted becomes obsessed with imagined abnormalities of the body (e.g., thinking that his or her body is overweight, deformed, or not muscular enough, etc.) as mentioned in Phillips et al. [12], Previously found gender differences in BDD will help us map this disorder to our social media study of the youth in the MENA region. Indeed, BDD is associated with risky psychological problems such as anxiety, low self-esteem, depression, eating disorders, and substance use. Moreover, stigmatization via social media of psychological problems can impede youths from seeking treatment either prior to or after engaging in risky behaviors, as explained in Clement et al. [13]. We plan to investigate how stigma might play a role in risky behaviors and hinder help-seeking. To the best of our knowledge, our study is novel as it is the only study of its kind covering the MENA region. The risky youth behaviors in the MENA region will be analyzed from a social media perspective using a combination of corpus linguistics and natural language processing methods combined with computational social science methods.

3. Research Issues and our Approach

In order to build our corpus, we plan to collect and annotate publicly available historical social media data from 10 sub-regions in the Middle East and North Africa to create a balanced dataset covering the depression and self-harm issues. The dataset will be covering the standard Arabic as well as the various Arabic dialects including the code-switching and the transliterated Arabic forms.

Once the data set is collected, cleaned and annotated, a linguistic analysis and a behavioral study of the corpus will be performed to identify the existence of linguistics cues related to depressive messages. This will help answer several key questions related to the root causes, the role of social media, and the sentiments expressed by the youth when it comes to these risky behaviors. An analysis of a historical timeline of a given subject might reveal common cues, or indicators that may help the stakeholders predict the occurrence of such behavior, and potentially detect relevant signs early on, to allow for effective intervention. We plan also to build prediction models and tools based on the collected data. Such models and tools could hold significant value in assisting the youth themselves, as well as social support services, and the wider community in the MENA region in general.

From a computational social analysis perspective, a set of classifiers would be built for demographic predictions to understand the background and features of the social media users who get identified with potential risky behavior. The methodologies to be used include machine learning approaches such as support vector machines (SVMs), deep learning (DNNs), and others. In addition, topic detection algorithms will be applied to learn the interests of users as part of their profiling. All the information to be extracted about users will then be used to build a prediction model to identify users or actions that can be potentially risky in the future.

3.1. Data Collection and Annotation

In order to collect our data, a list of accounts of holders who are tweeting topics related to depression and self-harm would be identified. An expected list of thousands of accounts are to be identified. All profile, network, and timeline information of these accounts are to be crawled. Twitter API allows the crawling of the latest 3,200 tweets of a given user account, besides the account's profile information (such as location, description, and profile image) and the network information including the list of followers and followees (those accounts a user follows). All the above information would be crawled in this task and saved in a highly structured database. The expected amount of time for crawling such collection would be 4-6 months. This includes hundreds of millions of tweets and network information. Once the historic information of accounts is downloaded, another API will be used to lively follow any new posts from these accounts and keep collecting and saving them over time.

We plan to analyze the discourse patterns of the youth in the MENA region, particularly the sentiments expressed while posting messages on social media that provides insights into depressive behavior. This will proceed in two steps. In the first step, we will use MADAMIRA [14], a system for morphological analysis and disambiguation of Arabic and its dialects to process the corpus and generate linguistic information such as tokenization, diacritization, lemmatization, part-of-speech tagging for the whole dataset. Later on, we will generate the most frequent lexical expressions related to depression and self-harm according to the observed linguistic patterns.

In the next step, we will conduct sentiment analysis as explained by Pang et al. [15]. This will help us in automatically judging if the sentiment of a given text (e.g., a tweet) is positive, negative, or neutral. Later on, we will analyze the emotions expressed in a positive or negative tweet based on Ekman's six emotions - joy, sadness, anger, fear, disgust, and surprise [16]. We will refer to the Arabic translation of the NRC emotion lexicon [17] to detect terms conveying emotions. We will also expand the NRC emotion lexicon with new emotion terms and phrases as expressed in the different Arabic dialects. Regarding the computational social analysis part, a set of classifiers would be built for demographic predictions to understand the background and features of the social media users who get identified with potentially risky behavior. The methodologies to be used includes machine learning approaches such as support vector machines (SVMs), deep learning (DNNs), and others. In addition, topic detection algorithms will be applied to learn the interest of users as part of their profiling. All the information to be extracted about users will be then used for building a prediction model to identify users or actions that can be potentially risky in the future.

From the psycho-social-behavioral aspect, the corpus annotation task will include the annotation of the types of emojis used to emphasize/enhance text and the types of links to images, videos, vlogs, blogs, etc. that are included. We will conduct frequency analyses by group membership, that being gender, age, ethnicity, location, socioeconomic status, etc. Moreover, we plan to conduct correlational analyses to determine what types of codes or other information might be predictive of depression of the self-harm. What types of linguistic markers are associated with youth depression or self-harm? Does group membership moderate these behaviors? For example, are males more likely to likely to show depression symptoms? Further, we will also examine, to the extent that data are available, whether there are correlations between other risk factors such as family instability (e.g., divorce), learning disabilities (e.g., dyslexia, ADHD), irregular sleep patterns (e.g., sleeping too much, sleeping too little, sleeping at odd hours), body dysmorphia, cyberbullying, peer pressure, familial pressure, cultural pressure, and excessive Internet usage.

3.2. Tools

With the previous steps, a large set of data and insights would be generated. Deep understanding of the main features affecting youths' depression will be clear. In this task, the knowledge constructed from previous steps would be used to train models by using different machine learning approaches (e.g., SVMs and Neural Networks) to be applied to the Arabic Twitter stream for potential detection of future risky behavior. This part of the work will be evaluated using precision and recall. The system can be tuned to achieve higher precision than recall in automated actions. However, for some aspects, we will be interested in a higher recall over precision, especially for high risk behaviour such as suicide and self-harm.

4. Conclusion

Social media has become a primary online resource to study various health issues such as depression and self-harm. In this paper, we presented our approach to building a corpus and related tools to address the depression and self-harm issues as observed in social media in the Arab world. To the best of our knowledge, the corpus will be the first Arabic multi-dialect social media youth depression corpus. We believe that the annotated corpus to will be a valuable resource to be used by linguists, sociologists, computer scientists, psychologists, policy makers, etc.

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References

[1] Cooper, M. L. (2002). Alcohol use and risky sexual behavior among college students and youth: evaluating the evidence. Journal of Studies on Alcohol, supplement, (14), 101-117.

[2] Zafarani, R., Liu, H. (2013). Connecting Users across Social Media Sites: A Behavioral-Modeling Approach, In Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining (pp. 41-49). ACM.

[3] Zafarani, R., Abbasi, MA., Liu, H. (2014). Social Media Mining: An Introduction, Cambridge University Press, 2014.

[4] Pennebaker James W. (2011). The secret life of pronouns. New Scientist, 211(2828):42–45.

[5] Pennebaker James W., Cindy K. Chung, Molly Ireland, Amy Gonzales, and Roger J. Booth (2007). The development and psychometric properties of LIWC2007. LIWC.net, Austin, TX.

[6] Ybarra, M. L., & Mitchell, K. J. (2008). How risky are social networking sites? A comparison of places online where youth sexual solicitation and harassment occurs. Pediatrics, 121(2), e350-e357.

[7] Wolak, J., Finkelhor, D., & Mitchell, K. (2008). Is talking online to unknown people always risky? Distinguishing online interaction styles in a national sample of youth Internet users. CyberPsychology & Behavior, 11(3), 340-343.

[8] De Choudhury, M., Counts, S., & Horvitz, E. (2013a). Social media as a measurement tool of depression in populations. In Proceedings of the 5th Annual ACM Web Science Conference (pp. 47-56). ACM.

[9] De Choudhury, M., Gamon, M., Counts, S., & Horvitz, E. (2013b). Predicting Depression via Social Media. In ICWSM (p. 2).

[10] Weber, A. S. (2013). Youth unemployment and sustainable development: Case study of Qatar. Revista de Asistenţă Socială, (1), 47-57.

[11] Abokhodair, N., Abbar, S., Vieweg, S., & Mejova, Y. (2016, May). Privacy and twitter in Qatar: traditional values in the digital world. In Proceedings of the 8th ACM Conference on Web Science (pp. 66-77). ACM.

[12] Phillips, K. A., Menard, W., & Fay, C. (2006). Gender similarities and differences in 200 individuals with body dysmorphic disorder. Comprehensive Psychiatry, 47(2), 77-87.

[13] Clement, S., Schauman, O., Graham, T., Maggioni, F., Evans-Lacko, S., Bezborodovs, N. & Thornicroft, G. (2015). What is the impact of mental health-related stigma on help-seeking? A systematic review of quantitative and qualitative studies. Psychological Medicine, 45(01), 11-27.

[14] Arfath Pasha, Mohamed Al-Badrashiny, Mona Diab, Ahmed El Kholy, Ramy Eskander, Nizar Habash,

Manoj Pooleery, Owen Rambow, and Ryan M. Roth (2014). MADAMIRA: A Fast, Comprehensive Tool for Morphological Analysis and Disambiguation of Arabic LREC, 2014.

[15] Pang, Bo; Lee, Lillian; Vaithyanathan, Shivakumar (2002). "Thumbs up? Sentiment Classification using Machine Learning Techniques". Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 79–86.

[16] Ekman Paul (1992). An argument for basic emotions. Cognition and Emotion, 6(3):169–200.

[17] Mohammad Saif and Peter Turney (2010). Emotions Evoked by Common Words and Phrases: Using Mechanical Turk to Create an Emotion Lexicon. In Proceedings of the NAACL-HLT 2010 Workshop on Computational Approaches to Analysis and Generation of Emotion in Text, June 2010, LA, California.